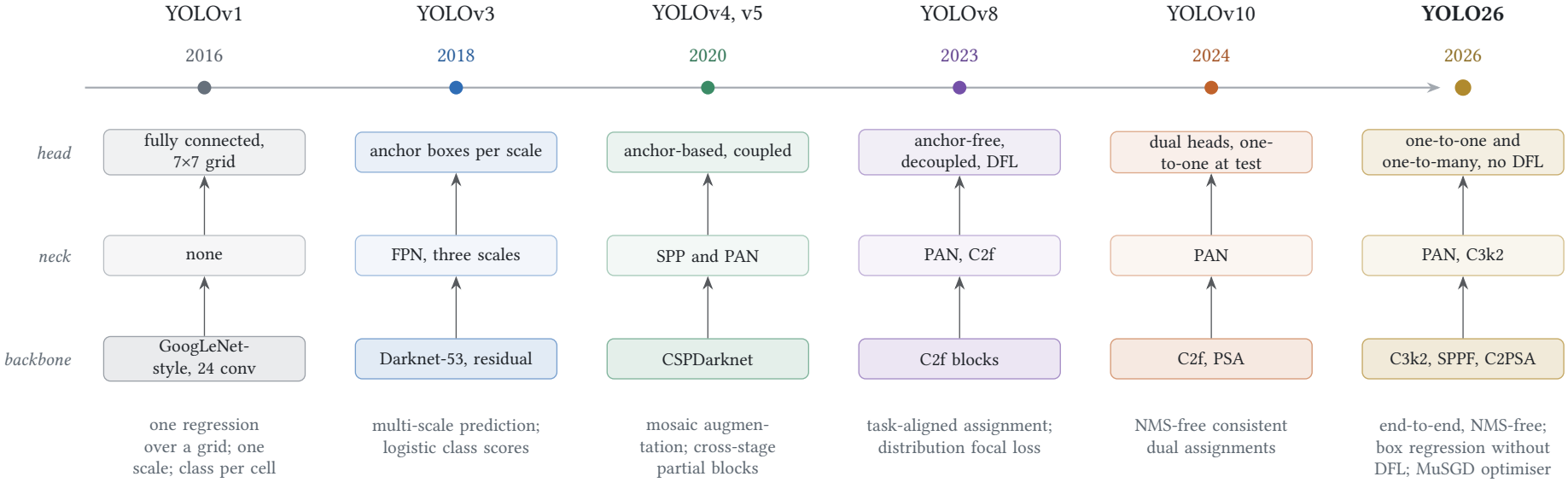
YOLOv1
2016
YOLOv3
2018
YOLOv4, v5
2020
YOLOv8
2023
YOLOv10
2024
YOLO26
2026
head
neck
backbone
fully connected, 7×7 grid
none
GoogLeNet-style, 24 conv
anchor boxes per scale
FPN, three scales
Darknet-53, residual
anchor-based, coupled
SPP and PAN
CSPDarknet
anchor-free, decoupled, DFL
PAN, C2f
C2f blocks
dual heads, one-to-one at test
PAN
C2f, PSA
one-to-one and one-to-many, no DFL
PAN, C3k2
C3k2, SPPF, C2PSA
one regression over a grid; one scale; class per cell
multi-scale prediction; logistic class scores
mosaic augmentation; cross-stage partial blocks
task-aligned assignment; distribution focal loss
NMS-free consistent dual assignments
end-to-end, NMS-free; box regression without DFL; MuSGD optimiser